

Detailed Solutions

SXC-2026-UG-6-P-12

Physics (MJ-12)

Group–A

1. Answer the following questions.

[1 × 5 = 5 marks]

(a) Define expectation value of an observable.

In quantum mechanics an observable is represented by a Hermitian operator, say $\hat{A}$. If the system is in a normalized state ψ , then the expectation value of the observable is the average value obtained from a large number of identical measurements:

$$\langle A \rangle = \int \psi^*(\mathbf{r}, t) \hat{A} \psi(\mathbf{r}, t) d\tau.$$

For a one-dimensional system this becomes

$$\langle A \rangle = \int_{-\infty}^{\infty} \psi^*(x, t) \hat{A} \psi(x, t) dx.$$

Thus the expectation value is the mean measured value of an observable in a given quantum state.

(b) Write the possible values of magnetic quantum number m for $l = 3$.

For a given orbital quantum number l , the magnetic quantum number can take the integral values

$$m = -l, -l + 1, \dots, 0, \dots, l - 1, l.$$

Therefore, for $l = 3$,

$$m = -3, -2, -1, 0, +1, +2, +3.$$

The possible values are $m = -3, -2, -1, 0, 1, 2, 3$.

(c) State the selection rules for Zeeman effect.

For electric dipole transitions observed in the Zeeman effect, the main selection rules are

$$\Delta l = \pm 1, \quad \Delta m = 0, \pm 1.$$

In the anomalous Zeeman effect, where total angular momentum j is used, the corresponding rule is

$$\Delta j = 0, \pm 1, \quad j = 0 \not\leftrightarrow j = 0.$$

The components with $\Delta m = 0$ are called π components, and those with $\Delta m = \pm 1$ are called σ components.

The essential Zeeman selection rules are $\Delta l = \pm 1$ and $\Delta m = 0, \pm 1$; for fine-structure levels $\Delta j = 0, \pm 1$, except $0 \rightarrow 0$.

(d) Write the ground state energy of a particle in a one-dimensional infinite potential well.

For a particle of mass m in a one-dimensional infinite well of width L , the allowed energies are

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2} = \frac{n^2 h^2}{8mL^2}, \quad n = 1, 2, 3, \dots$$

The ground state corresponds to $n = 1$. Hence

$$E_1 = \frac{\pi^2 \hbar^2}{2mL^2} = \frac{h^2}{8mL^2}.$$

The ground state energy is $E_1 = \frac{\pi^2 \hbar^2}{2mL^2} = \frac{h^2}{8mL^2}$.

(e) Write the uncertainty principle between position and momentum.

The Heisenberg uncertainty relation between position and the corresponding component of momentum is

$$\Delta x \Delta p_x \geq \frac{\hbar}{2}.$$

It means that the position and momentum of a particle along the same direction cannot both be measured with arbitrary precision at the same time.

The position-momentum uncertainty principle is $\Delta x \Delta p_x \geq \hbar/2$.

2. A system is described by an initial wave function

$$\psi(x, 0) = c_1 \psi_1(x) + c_2 \psi_2(x),$$

where ψ_1 and ψ_2 are stationary states with energies E_1 and E_2 . Obtain the time-dependent wave function at any later time t . [5 marks]

For a time-independent Hamiltonian, a stationary state $\psi_n(x)$ with energy E_n evolves only by a phase factor:

$$\psi_n(x, t) = \psi_n(x) \exp\left(-\frac{iE_n t}{\hbar}\right).$$

Since the Schrödinger equation is linear, a superposition at $t = 0$ remains a superposition at later time. Therefore,

$$\psi(x, t) = c_1 \psi_1(x) \exp\left(-\frac{iE_1 t}{\hbar}\right) + c_2 \psi_2(x) \exp\left(-\frac{iE_2 t}{\hbar}\right).$$

If the state is normalized and ψ_1, ψ_2 are orthonormal, then

$$|c_1|^2 + |c_2|^2 = 1.$$

The probabilities of obtaining E_1 and E_2 are $|c_1|^2$ and $|c_2|^2$, respectively, and these do not change with time. What changes is the relative phase:

$$\exp\left[-\frac{i(E_2 - E_1)t}{\hbar}\right].$$

The required time-dependent wave function is

$$\psi(x, t) = c_1 \psi_1(x) e^{-iE_1 t/\hbar} + c_2 \psi_2(x) e^{-iE_2 t/\hbar}.$$

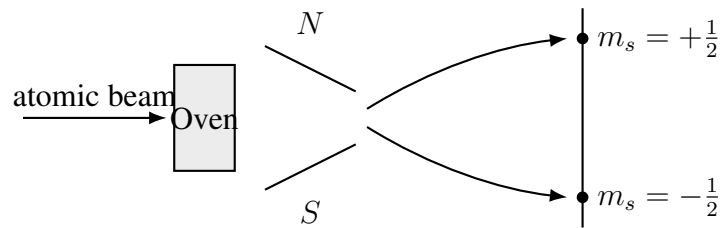
3. Explain how the Stern-Gerlach experiment provides evidence for electron spin and magnetic moment. [5 marks]

The Stern-Gerlach experiment sends a beam of neutral atoms through a non-uniform magnetic field. A magnetic dipole moment μ in a non-uniform field experiences a force

$$\mathbf{F} = \nabla(\mu \cdot \mathbf{B}).$$

If the magnetic field gradient is along the z -direction, the force is approximately

$$F_z = \mu_z \frac{\partial B_z}{\partial z}.$$



Classically, if the magnetic moment could point in any direction, a continuous smear should appear on the detector. Instead, the beam splits into discrete components. In the usual silver-atom version, the orbital angular momentum of the outer electron is zero ($l = 0$), so the observed splitting cannot be due to orbital angular momentum. It is explained by intrinsic electron spin.

For an electron,

$$s = \frac{1}{2}, \quad m_s = \pm \frac{1}{2},$$

so the spin angular momentum has two allowed projections:

$$S_z = m_s \hbar = \pm \frac{\hbar}{2}.$$

The spin magnetic moment is

$$\mu_s = -g_s \mu_B \frac{\mathbf{S}}{\hbar}, \quad g_s \simeq 2,$$

where $\mu_B = e\hbar/(2m_e)$ is the Bohr magneton. Hence the two allowed values of S_z correspond to two allowed magnetic-moment projections, producing two deflected beams.

The Stern-Gerlach experiment proves space quantization and gives direct evidence that the electron has intrinsic spin $s = 1/2$ and an associated magnetic moment.

Group-B

Answer any four questions of the following in the examination. Detailed solutions to all six are given here. [15 × 4 = 60 marks]

4. Construct a Gaussian wave packet for a free particle in one-dimension. Obtain its time evolution, show that it spreads with time and relate this spreading to the position-momentum uncertainty principle. [15 marks]

A free particle in one dimension has Hamiltonian

$$\hat{H} = \frac{\hat{p}^2}{2m}.$$

A plane wave e^{ikx} has energy

$$E = \frac{\hbar^2 k^2}{2m}, \quad \omega(k) = \frac{\hbar k^2}{2m}.$$

A localized particle is formed by superposing many plane waves. A convenient minimum-uncertainty packet at $t = 0$ is

$$\psi(x, 0) = \left(\frac{1}{2\pi\sigma_0^2} \right)^{1/4} \exp \left[-\frac{(x - x_0)^2}{4\sigma_0^2} + ik_0 x \right],$$

where x_0 is the initial mean position, k_0 is the mean wave number, and σ_0 is the initial position uncertainty:

$$\Delta x(0) = \sigma_0.$$

Equivalently, the packet can be written as a Fourier superposition,

$$\psi(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi(k, 0) \exp [ikx - i\omega(k)t] dk,$$

where $\phi(k, 0)$ is sharply peaked around k_0 .

For the Gaussian packet above, the exact time-evolved wave function is

$$\psi(x, t) = \left(\frac{1}{2\pi\sigma_0^2} \right)^{1/4} \frac{1}{\sqrt{1 + i\hbar t/(2m\sigma_0^2)}} \exp \left[-\frac{(x - x_0 - v_g t)^2}{4\sigma_0^2 (1 + i\hbar t/(2m\sigma_0^2))} + ik_0 x - \frac{i\hbar k_0^2 t}{2m} \right],$$

where the group velocity is

$$v_g = \left. \frac{d\omega}{dk} \right|_{k=k_0} = \frac{\hbar k_0}{m}.$$

Thus the centre of the packet moves as

$$\langle x \rangle = x_0 + v_g t.$$

The probability density is obtained by taking the modulus squared:

$$|\psi(x, t)|^2 = \frac{1}{\sqrt{2\pi} \sigma_t} \exp \left[-\frac{(x - x_0 - v_g t)^2}{2\sigma_t^2} \right],$$

where

$$\sigma_t = \sigma_0 \sqrt{1 + \left(\frac{\hbar t}{2m\sigma_0^2} \right)^2}.$$

Therefore

$$\Delta x(t) = \sigma_t = \sigma_0 \sqrt{1 + \left(\frac{\hbar t}{2m\sigma_0^2} \right)^2}.$$

Since the particle is free, its momentum distribution does not change with time:

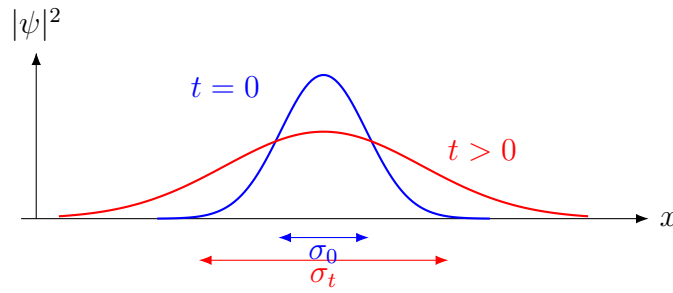
$$\Delta p = \frac{\hbar}{2\sigma_0}.$$

Hence

$$\Delta x(t) \Delta p = \frac{\hbar}{2} \sqrt{1 + \left(\frac{\hbar t}{2m\sigma_0^2} \right)^2} \geq \frac{\hbar}{2}.$$

At $t = 0$ the Gaussian is a minimum-uncertainty state, but for $t > 0$ the packet spreads because different wave-number components have different velocities:

$$v(k) = \frac{\hbar k}{m}.$$



A free Gaussian wave packet remains Gaussian, its centre moves with group velocity $\hbar k_0/m$, and its width increases as

$$\Delta x(t) = \sigma_0 \sqrt{1 + \left(\frac{\hbar t}{2m\sigma_0^2} \right)^2}.$$

The spreading is a direct consequence of the momentum spread required by $\Delta x \Delta p \geq \hbar/2$.

5. Obtain the energy eigenvalues and eigenfunctions of a one-dimensional quantum harmonic oscillator using the Frobenius method. Discuss the role of Hermite polynomials in the solution. [15 marks]

For a one-dimensional harmonic oscillator,

$$V(x) = \frac{1}{2}m\omega^2 x^2.$$

The time-independent Schrödinger equation is

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + \frac{1}{2}m\omega^2 x^2 \psi = E\psi.$$

Introduce the dimensionless variable

$$\xi = \sqrt{\frac{m\omega}{\hbar}} x$$

and the dimensionless energy parameter

$$\epsilon = \frac{2E}{\hbar\omega}.$$

Then the Schrödinger equation becomes

$$\frac{d^2\psi}{d\xi^2} + (\epsilon - \xi^2)\psi = 0.$$

For large $|\xi|$, the dominant equation is approximately

$$\frac{d^2\psi}{d\xi^2} - \xi^2\psi \simeq 0,$$

whose physically acceptable behaviour is

$$\psi \sim e^{-\xi^2/2}.$$

So we write

$$\psi(\xi) = H(\xi)e^{-\xi^2/2}.$$

Substitution gives

$$\frac{d^2H}{d\xi^2} - 2\xi \frac{dH}{d\xi} + (\epsilon - 1)H = 0.$$

Now use a Frobenius power series

$$H(\xi) = \sum_{r=0}^{\infty} a_r \xi^r.$$

Then

$$H' = \sum_{r=1}^{\infty} r a_r \xi^{r-1}, \quad H'' = \sum_{r=2}^{\infty} r(r-1) a_r \xi^{r-2}.$$

After shifting indices and equating the coefficient of each power of ξ to zero, we get the recurrence relation

$$a_{r+2} = \frac{2r+1-\epsilon}{(r+2)(r+1)} a_r.$$

This recurrence produces two independent series: one even and one odd. If the series does not terminate, then for large r it behaves like e^{ξ^2} , making

$$\psi = H(\xi) e^{-\xi^2/2}$$

non-normalizable. Therefore the series must terminate. Termination occurs when

$$2n+1-\epsilon=0,$$

where $n=0, 1, 2, \dots$. Thus

$$\epsilon = 2n+1.$$

Since $\epsilon = 2E/(\hbar\omega)$,

$$E_n = \left(n + \frac{1}{2}\right) \hbar\omega, \quad n = 0, 1, 2, \dots$$

For these allowed values of energy, the terminating polynomial $H(\xi)$ is the Hermite polynomial $H_n(\xi)$. Therefore the normalized eigenfunctions are

$$\psi_n(x) = \left(\frac{\alpha}{\sqrt{\pi} 2^n n!}\right)^{1/2} H_n(\alpha x) \exp\left(-\frac{\alpha^2 x^2}{2}\right), \quad \alpha = \sqrt{\frac{m\omega}{\hbar}}.$$

The first few Hermite polynomials are

$$H_0(\xi) = 1, \quad H_1(\xi) = 2\xi, \quad H_2(\xi) = 4\xi^2 - 2.$$

Hermite polynomials are important because they are precisely the polynomial factors that make the oscillator wave functions finite and normalizable. Their order n labels the energy level. They also determine the parity of the wave function: H_n is even for even n and odd for odd n .

The oscillator energies are

$$E_n = \left(n + \frac{1}{2}\right) \hbar\omega, \quad n = 0, 1, 2, \dots$$

and the eigenfunctions are Gaussian factors multiplied by Hermite polynomials:

$$\psi_n(x) = \left(\frac{\alpha}{\sqrt{\pi} 2^n n!}\right)^{1/2} H_n(\alpha x) e^{-\alpha^2 x^2/2}.$$

6. Define the angular momentum operators and derive their commutation relations. Set up the time-independent Schrödinger equation in spherical polar coordinates and solve it by separation of variables. [15 marks]

The orbital angular momentum operator is defined by

$$\mathbf{L} = \mathbf{r} \times \mathbf{p}, \quad \mathbf{p} = -i\hbar\nabla.$$

Its Cartesian components are

$$L_x = yp_z - zp_y, \quad L_y = zp_x - xp_z, \quad L_z = xp_y - yp_x.$$

Using the canonical commutation relations

$$[x_i, p_j] = i\hbar\delta_{ij}, \quad [x_i, x_j] = [p_i, p_j] = 0,$$

one obtains

$$[L_x, L_y] = i\hbar L_z, \quad [L_y, L_z] = i\hbar L_x, \quad [L_z, L_x] = i\hbar L_y.$$

In compact notation,

$$[L_i, L_j] = i\hbar\epsilon_{ijk}L_k.$$

Also,

$$L^2 = L_x^2 + L_y^2 + L_z^2,$$

and

$$[L^2, L_x] = [L^2, L_y] = [L^2, L_z] = 0.$$

Thus L^2 and one component, conventionally L_z , can be simultaneously specified.

In spherical polar coordinates,

$$L_z = -i\hbar\frac{\partial}{\partial\phi},$$

and

$$L^2 = -\hbar^2 \left[\frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \left(\sin\theta \frac{\partial}{\partial\theta} \right) + \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial\phi^2} \right].$$

For a central potential $V(r)$, the time-independent Schrödinger equation is

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V(r) \right] \psi(r, \theta, \phi) = E\psi(r, \theta, \phi),$$

where

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin\theta} \frac{\partial}{\partial\theta} \left(\sin\theta \frac{\partial}{\partial\theta} \right) + \frac{1}{r^2 \sin^2\theta} \frac{\partial^2}{\partial\phi^2}.$$

Equivalently,

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) - \frac{L^2}{\hbar^2 r^2}.$$

Hence

$$\left[-\frac{\hbar^2}{2m} \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{L^2}{2mr^2} + V(r) \right] \psi = E\psi.$$

Now take a separated solution

$$\psi(r, \theta, \phi) = R(r)Y(\theta, \phi) = R(r)\Theta(\theta)\Phi(\phi).$$

The angular part satisfies

$$L^2 Y_l^m(\theta, \phi) = l(l+1)\hbar^2 Y_l^m(\theta, \phi),$$

and

$$L_z Y_l^m(\theta, \phi) = m\hbar Y_l^m(\theta, \phi).$$

The azimuthal equation gives

$$\frac{d^2\Phi}{d\phi^2} + m^2\Phi = 0,$$

so

$$\Phi(\phi) = e^{im\phi}.$$

Single-valuedness requires

$$m = 0, \pm 1, \pm 2, \dots$$

The polar equation gives the associated Legendre functions, and the allowed angular solutions are the spherical harmonics

$$Y_l^m(\theta, \phi) = N_{lm} P_l^m(\cos \theta) e^{im\phi},$$

where

$$l = 0, 1, 2, \dots, \quad m = -l, -l+1, \dots, l.$$

The radial equation is

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \left[\frac{2m}{\hbar^2} (E - V(r)) - \frac{l(l+1)}{r^2} \right] R = 0.$$

It is often simplified by putting $u(r) = rR(r)$:

$$\frac{d^2u}{dr^2} + \left[\frac{2m}{\hbar^2} (E - V(r)) - \frac{l(l+1)}{r^2} \right] u = 0.$$

The angular momentum operators satisfy $[L_i, L_j] = i\hbar\epsilon_{ijk}L_k$ and $[L^2, L_i] = 0$. In a central potential, separation gives angular eigenfunctions $Y_l^m(\theta, \phi)$ and a radial equation with the effective centrifugal term $l(l+1)\hbar^2/(2mr^2)$.

7. Discuss electron spin and spin angular momentum and derive Larmor's theorem for a charged particle in a magnetic field. [15 marks]

Electron spin is an intrinsic angular momentum. It is not due to literal rotation of the electron as a small charged sphere. For an electron,

$$s = \frac{1}{2}.$$

The spin angular momentum operators satisfy the same algebra as orbital angular momentum:

$$[S_i, S_j] = i\hbar\epsilon_{ijk}S_k.$$

The simultaneous eigenvalue equations are

$$S^2 |s, m_s\rangle = s(s+1)\hbar^2 |s, m_s\rangle,$$

$$S_z |s, m_s\rangle = m_s\hbar |s, m_s\rangle.$$

For the electron,

$$s = \frac{1}{2}, \quad m_s = \pm \frac{1}{2}.$$

Therefore

$$S^2 = \frac{3}{4}\hbar^2, \quad S_z = \pm \frac{\hbar}{2}.$$

The spin magnetic moment is

$$\boldsymbol{\mu}_s = -g_s \mu_B \frac{\mathbf{S}}{\hbar}, \quad g_s \simeq 2,$$

where

$$\mu_B = \frac{e\hbar}{2m_e}$$

is the Bohr magneton. The negative sign appears because the electron has negative charge.

For a charged particle of charge q and mass m moving in a central potential in a weak uniform magnetic field $\mathbf{B}$, choose the vector potential

$$\mathbf{A} = \frac{1}{2}\mathbf{B} \times \mathbf{r}.$$

The Hamiltonian is

$$H = \frac{1}{2m}(\mathbf{p} - q\mathbf{A})^2 + V(r).$$

Keeping only terms first order in B ,

$$H \simeq \frac{p^2}{2m} + V(r) - \frac{q}{2m}\mathbf{B} \cdot \mathbf{L}.$$

The magnetic interaction energy can also be written

$$H' = -\boldsymbol{\mu}_L \cdot \mathbf{B}, \quad \boldsymbol{\mu}_L = \frac{q}{2m}\mathbf{L}.$$

The torque on the magnetic moment is

$$\boldsymbol{\tau} = \boldsymbol{\mu}_L \times \mathbf{B}.$$

Since torque equals the rate of change of angular momentum,

$$\frac{d\mathbf{L}}{dt} = \boldsymbol{\mu}_L \times \mathbf{B} = \frac{q}{2m}\mathbf{L} \times \mathbf{B}.$$

This equation describes precession of $\mathbf{L}$ about the magnetic field. It may be written in the form

$$\frac{d\mathbf{L}}{dt} = \boldsymbol{\omega}_L \times \mathbf{L},$$

where

$$\boldsymbol{\omega}_L = -\frac{q}{2m}\mathbf{B}.$$

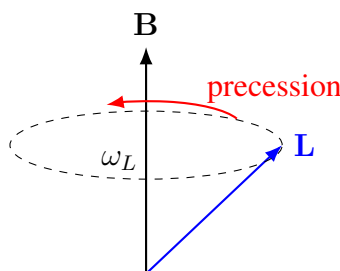
Thus the magnitude of the Larmor angular frequency is

$$\omega_L = \frac{|q|B}{2m}.$$

For an electron, $q = -e$, so

$$\omega_L = \frac{eB}{2m_e}.$$

Larmor's theorem states that, to first order in a weak uniform magnetic field, the effect of the magnetic field on the orbital motion of a charged particle is equivalent to a uniform precession of the orbit about the magnetic field direction with angular frequency $\omega_L = |q|B/(2m)$.



Electron spin has $s = 1/2$ and two projections $m_s = \pm 1/2$. A charged particle in a weak uniform magnetic field undergoes Larmor precession with angular frequency $\omega_L = |q|\hbar/(2m)$; for an electron this is $e\hbar/(2m_e)$.

8. Explain the Paschen-Back effect and the Stark effect qualitatively. Discuss their physical origin and describe how atomic spectral lines are affected in the presence of strong magnetic and external electric fields. [15 marks]

Paschen-Back effect. The Paschen-Back effect is the splitting of atomic energy levels and spectral lines in a strong magnetic field. In weak magnetic fields the spin-orbit interaction remains important, so $\mathbf{L}$ and $\mathbf{S}$ combine to give

$$\mathbf{J} = \mathbf{L} + \mathbf{S}.$$

The levels are then labelled by J and m_J , giving the usual anomalous Zeeman effect.

In a strong magnetic field, however, the magnetic interaction becomes comparable to or larger than the spin-orbit interaction. Then $\mathbf{L}$ and $\mathbf{S}$ are no longer strongly coupled to each other; instead they precess almost independently about the external magnetic field. The good quantum numbers become approximately m_l and m_s , rather than J and m_J .

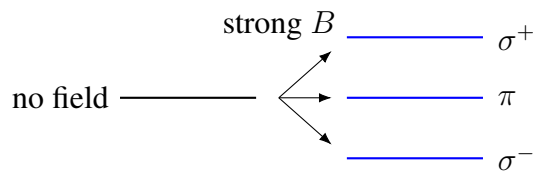
The magnetic energy shift in the strong-field limit is approximately

$$\Delta E = \mu_B B(m_l + g_s m_s),$$

with $g_s \simeq 2$. Since the electric dipole transition does not change spin, the selection rules are approximately

$$\Delta m_l = 0, \pm 1, \quad \Delta m_s = 0.$$

Thus the complicated anomalous Zeeman pattern tends toward a simpler strong-field pattern.



Stark effect. The Stark effect is the splitting and shifting of atomic spectral lines in an external electric field. The physical origin is the interaction between the atom's electric dipole moment and the external electric field. If the electric field is along the z -axis, the perturbation is

$$H' = -\mathbf{d} \cdot \mathbf{E}.$$

For an electron in an atom, this is often written as

$$H' = eEz,$$

up to the usual sign convention for the electron charge.

There are two important cases.

i. Linear Stark effect: In atoms with degenerate states of opposite parity, such as hydrogen, the first-order energy shift may be proportional to E :

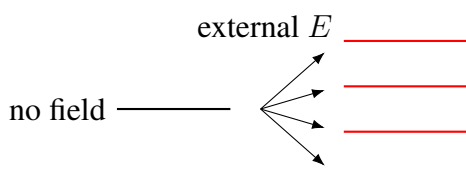
$$\Delta E \propto E.$$

ii. Quadratic Stark effect: In non-degenerate states, the first-order shift often vanishes by parity, and the leading shift is second order:

$$\Delta E = -\frac{1}{2}\alpha E^2,$$

where α is the polarizability.

The electric field removes degeneracy, mixes states of different parity, and causes spectral lines to split into several components or shift from their original positions.



The Paschen-Back effect is strong-field magnetic splitting where L and S become effectively decoupled. The Stark effect is electric-field splitting or shifting caused by the dipole interaction $-\mathbf{d} \cdot \mathbf{E}$. Both effects reveal how external fields remove degeneracies and modify atomic spectral lines.

9. Explain the spectral notation for atomic states. For an electronic configuration p^2 , determine the possible term symbols using L-S coupling and apply Hund's rules to identify the ground state. [15 marks]

In L-S coupling, also called Russell-Saunders coupling, the orbital angular momenta of individual electrons combine to give total orbital angular momentum

$$\mathbf{L} = \sum_i \mathbf{l}_i,$$

and the spins combine to give total spin angular momentum

$$\mathbf{S} = \sum_i \mathbf{s}_i.$$

Then

$$\mathbf{J} = \mathbf{L} + \mathbf{S}.$$

The atomic term symbol is written as

$$^{2S+1}L_J.$$

Here:

- $2S + 1$ is the spin multiplicity.
- L is written using letters:

$$L = 0, 1, 2, 3, 4, \dots \Rightarrow S, P, D, F, G, \dots$$

- J is the total angular momentum quantum number:

$$J = L + S, L + S - 1, \dots, |L - S|.$$

For a p^2 configuration, each electron has

$$l = 1, \quad s = \frac{1}{2}.$$

Combining two p -electron orbital angular momenta gives possible values

$$L = 0, 1, 2,$$

corresponding to

$$S, P, D.$$

Combining the two spins gives

$$S = 0 \quad \text{or} \quad S = 1.$$

At first sight this suggests singlet and triplet terms. However, because the two electrons are equivalent electrons in the same p subshell, the Pauli exclusion principle removes some combinations. The allowed L-S terms for p^2 are

$$^1S, \quad ^1D, \quad ^3P.$$

Counting microstates checks the result:

$$^1S : 1, \quad ^1D : 5, \quad ^3P : 3(2L + 1) = 3 \times 3 = 9.$$

Thus

$$1 + 5 + 9 = 15,$$

which equals the number of possible p^2 microstates:

$$\binom{6}{2} = 15.$$

Now include the possible J values.

$$^1S : \quad S = 0, L = 0 \quad \Rightarrow \quad J = 0,$$

so

$$^1S_0.$$

$$^1D : \quad S = 0, L = 2 \quad \Rightarrow \quad J = 2,$$

so

$$^1D_2.$$

$$^3P : \quad S = 1, L = 1 \quad \Rightarrow \quad J = 2, 1, 0,$$

so

$$^3P_2, \quad ^3P_1, \quad ^3P_0.$$

Therefore the complete set of levels is

$$^1S_0, \quad ^1D_2, \quad ^3P_0, \quad ^3P_1, \quad ^3P_2.$$

Hund's rules now determine the ground state.

Rule 1: The term with maximum multiplicity lies lowest. Hence the triplet term 3P lies below the singlet terms.

Rule 2: For a given multiplicity, the term with larger L lies lower. This does not change the choice because the only triplet term here is 3P .

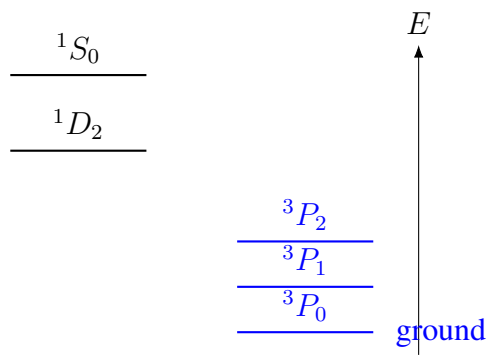
Rule 3: For a subshell less than half-filled, the level with the smallest J lies lowest; for a subshell more than half-filled, the largest J lies lowest.

The p subshell can hold six electrons. Since p^2 is less than half-filled, the smallest J in 3P_J is lowest:

$$J = 0.$$

Therefore the ground state is

$$^3P_0.$$



For p^2 , the allowed terms are 1S , 1D and 3P , giving the levels 1S_0 , 1D_2 , 3P_0 , 3P_1 , 3P_2 . Hund's rules give the ground state as 3P_0 .